Food Waste Risk Prediction

Project Plan | 6-Week Timeline

1. Project Overview

This project builds a machine-learning system that predicts the spoilage/waste risk level (LOW, MEDIUM, HIGH) of perishable products and recommends an action (normal selling, discount, or discount plus donation to an NGO). The goal is to reduce food waste and financial loss for retailers.

2. Dataset Summary

Item	Details
Files	food_waste_raw_1000.csv (original), food_waste_ml_1000.csv (scaled and encoded)
Size / Period	1,000 records, 1 Jan 2026 to 30 Jun 2026
Products (8)	Eggs, Rice, Bread, Chicken, Tomato, Banana, Apple, Milk
Features	quantity, price, expiry_days, temperature, day_of_week, is_weekend, predicted_demand, product
Target	risk_level: LOW 53.5%, MEDIUM 16.0%, HIGH 30.5%
Actions	Normal selling / Apply 10-15% discount / Apply 25% discount + Send to NGO

3. Objectives and Success Criteria

• Classify risk level with Macro F1 of at least 0.85 on a time-based test set.
• Achieve HIGH-risk recall of at least 0.90 (missing a high-risk item is the costliest error).
• Deliver an interpretable model (feature importance / SHAP) and a working demo dashboard.

4. Phases and Deliverables

Week	Phase	Key Tasks	Deliverable
1	Understanding and EDA	Define problem; analyze class balance, distributions, outliers (expiry_days ranges 1-89); check whether risk_level is a deterministic rule of expiry/quantity/demand (leakage check)	EDA notebook and findings summary
2	Data Preparation	Time-based train/test split; handle imbalance (class weights / SMOTE); feature engineering (quantity/demand ratio, month, product category)	Clean pipeline and feature set
3	Baseline Modeling	Logistic Regression, Decision Tree; establish baseline metrics	Baseline report
4	Advanced Modeling	Random Forest, XGBoost, LightGBM; cross-validation; hyperparameter tuning	Tuned model and comparison table
5	Interpretation and App	Feature importance, SHAP; map predictions to recommended actions; build Streamlit dashboard	Working demo
6	Finalization	Final report, presentation, documentation, clean GitHub repo	Final submission

5. Timeline (Gantt View)

Task	W1	W2	W3	W4	W5	W6
EDA						
Data Prep						
Baseline Models						
Advanced Models						
Interpretation + App						
Report + Delivery						

6. Tools and Technologies

Area	Tools
Language / Analysis	Python, pandas, NumPy, Matplotlib / Seaborn
Modeling	scikit-learn, XGBoost, LightGBM, imbalanced-learn
Explainability	SHAP
App / Delivery	Streamlit, Git / GitHub, Jupyter

7. Risks and Mitigation

Risk	Impact	Mitigation
Small dataset (1,000 rows), likely synthetic	Results may be overly optimistic and not generalize	Cross-validation, report limitations clearly, test on fresh data if available
Labels derived from fixed rules	Near-100% accuracy with no real value (data leakage)	Test rule-based baseline; remove or engineer features carefully; discuss openly
Class imbalance (MEDIUM only 16%)	Poor recall on minority class	Class weights, SMOTE, evaluate per-class metrics
Time leakage from random split	Inflated metrics	Chronological train/test split
Scope creep	Missed deadline	Fix MVP: model + simple dashboard; extras only if ahead of schedule

8. Evaluation Metrics

- **Primary:** Macro F1-score and HIGH-class recall.
- **Secondary:** per-class precision/recall, confusion matrix, ROC-AUC (one-vs-rest).
- **Business:** estimated waste reduction and value saved if recommended actions are followed.

9. Next Steps

- Run the EDA and leakage check on the dataset.
- Build the preprocessing pipeline and baseline model.
- Review results and decide on advanced models and dashboard scope.